

E-NJOY

E-NJOY is a student-built portable digital media device designed for reading, watching, listening, and managing personal content offline.

It is a device with touch screen in which I can read pdfs , Watch videos , open and play music and much more

I have decided to make this to solve the following problems :-

- 1. Watch or Read pdfs or maybe even play music when there's no Internet /.***
- 2. Help us to enjoy our time by watching limited [Our own uploads] E-NJOY is a student-built portable digital media device designed for reading, watching, listening, and managing personal content offline.***

It although as all good things have a disadvantage it needs to be charged too.

This is my Mockup :-



Well what else can I say U can see the mockup has the UI of the touch screen too , along with a few views of the device and an approx dimensional extent of the device.

Key Features

-  **E-Book & PDF Reader** — Read EPUBs, PDFs, and other digital documents on the touchscreen.
-  **Video Player** — Watch locally stored videos without needing a phone or laptop.
-  **Music Player** — Play locally stored music through built-in speakers, wired headphones, or Bluetooth audio.
-  **Image Gallery** — View personal images and other visual files.
-  **File Manager** — Browse, organize, copy, and delete files stored on the device.
-  **USB File Transfer** — Transfer personal content from USB storage or another device.
-  **Touchscreen Interface** — Navigate the device through a simple, dedicated touchscreen dashboard.
-  **Multiple Audio Options** — Support for built-in speakers, 3.5 mm wired headphones, and Bluetooth audio.
-  **Offline Use** — Access downloaded content without requiring a phone or constant internet connection.
-  **Portable Design** — Designed to run from a portable power source and be usable away from a desk.

***Note:** These are planned features for E-NJOY V1; the final implementation may change during prototyping.*

These are the lists of parts i would need to build it :-

<i>Part Number</i>	<i>Component</i>	<i>Purpose</i>	<i>Qty</i>
<i>1</i>	<i>Raspberry Pi 5</i>	<i>Main computer / runs E-NJOY software</i>	<i>1</i>
<i>2</i>	<i>7-inch touchscreen</i>	<i>Reading, video, music and UI</i>	<i>1</i>
<i>3</i>	<i>microSD card</i>	<i>Operating system and basic software</i>	<i>1</i>
<i>4</i>	<i>USB storage / SSD</i>	<i>Store ebooks, PDFs, videos, music and images</i>	<i>1</i>
<i>5</i>	<i>Speaker modules</i>	<i>Built-in audio</i>	<i>2</i>
<i>6</i>	<i>3.5 mm audio solution</i>	<i>Wired headphones</i>	<i>1</i>
<i>7</i>	<i>USB-C power solution</i>	<i>Power the device</i>	<i>1</i>
<i>8</i>	<i>Portable power bank</i>	<i>Portable operation</i>	<i>1</i>
<i>9</i>	<i>USB-A port/hub solution</i>	<i>File transfer and USB devices</i>	<i>1</i>
<i>10</i>	<i>Buttons / switches</i>	<i>Power and volume controls</i>	<i>3</i>
<i>11</i>	<i>Enclosure materials</i>	<i>Physical case</i>	<i>1</i>
<i>12</i>	<i>Cables/connectors</i>	<i>Internal connections</i>	<i>As required</i>

Our Estimated Budget

#	Component	Purpose	Estimated Cost
1	Raspberry Pi 5 (4GB)	Main computer	₹8,500
2	7" Touchscreen Display	Main interface	₹5,000
3	128GB microSD Card	OS + initial storage	₹900
4	Active Cooler for Raspberry Pi 5	Cooling	₹500
5	USB-C Power Supply	Main power	₹800
6	Portable Power Bank	Portable operation	₹1,200
7	Stereo Speakers	Built-in audio	₹500
8	USB Audio Adapter	3.5mm headphone support	₹400
9	USB Hub	File transfer / peripherals	₹500
10	Buttons & Switches	Physical controls	₹300
11	Cables & Connectors	Internal connections	₹700
12	Enclosure Materials	Device case/prototype	₹1,200
13	Mounting & Miscellaneous	Screws, brackets, adapters, etc.	₹500
—	TOTAL		₹23,000

Estimated V1 Budget: ₹23,000

The requested budget is intended to support the complete first prototype of E-NJOY, including the main computing board, touchscreen interface, storage, audio, portable power, connectivity, cooling, and enclosure materials. Components will primarily be sourced through Amazon India after funding approval. Prices may vary, so equivalent compatible components may be selected when necessary while remaining within the approved budget.

Software Plan

E-NJOY will run on Raspberry Pi OS with a touchscreen-friendly interface developed primarily in Python. The software will provide a simple dashboard for accessing books, PDFs, videos, music, images, and files.

E-NJOY is designed to work independently as a portable device while also allowing users to transfer their personal content from Windows, macOS, or Linux computers.

The initial software priority is reliability, simplicity, and offline access rather than adding unnecessary features.

File Transfer & Storage

Users will be able to transfer their own digital content to E-NJOY and access it locally on the device.

The initial version will support file transfer through USB storage and compatible connections. Future versions may introduce wireless file transfer through a local Wi-Fi connection.

E-NJOY will organize content into categories such as Books, Videos, Music, Images, and Documents.

Development Roadmap

Phase 1 – Design & Planning

Define the product, create hardware mockups, design the touchscreen interface, select components, and prepare the budget.

Phase 2 – Software Prototype

Develop the E-NJOY dashboard and basic file browsing, reading, and media playback functionality.

Phase 3 – Hardware Prototype

Assemble the Raspberry Pi 5, touchscreen, storage, audio system, and power system.

Phase 4 – Integration & Testing

Combine the software and hardware, test file transfer, media playback, touchscreen controls, audio, and portable operation.

Phase 5 – Final Prototype

Improve the enclosure, user interface, reliability, and overall user experience.

Current Status

E-NJOY is currently in the concept and design stage. The product concept, hardware requirements, interface ideas, parts list, and estimated budget have been defined. The next step is to begin software development and, after funding approval, construct and test the physical prototype.

Why This Project Matters

E-NJOY is designed to be more than a media player. It is a student-built project that combines software, electronics, product design, and problem-solving into one practical device.

The project will also provide an opportunity to learn real-world skills in Raspberry Pi development, Python, Linux, UI design, electronics, hardware integration, and prototyping.

With funding, I will be able to turn the concept into a working physical prototype, test it, document the development process, and share what I learn with other students and makers.

Funding Request

I am requesting funding of approximately ₹23,000 to build the first complete E-NJOY prototype.

The funding will primarily be used for the Raspberry Pi 5, touchscreen, storage, audio components, power system, cooling, connectivity, and enclosure materials described in the budget.

The goal is to use the funding responsibly to produce a functional prototype rather than simply purchase individual components without a clear build plan.

Thank You

Thank you for taking the time to review my E-NJOY project proposal.

I am excited to turn this idea into a real, working device and document the process so that other students and makers can learn from it.

Built by a student. Designed for students and makers.

— Mokshgna

E-NJOY

Your digital world. Anywhere.

Open Source • Student Built • Raspberry Pi 5